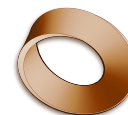


Forces 1 — Kinematics

AQA Combined Science: Trilogy 8464 • §6.5.4.1 • Physics 10 R



Seven lessons, Thu 3 September to the unit test on Tuesday 29 September 2026

Four periods a fortnight: **Week A** Thu P1 and Fri P1; **Week B** Tue P3 and Thu P2. Nine of Robert's forty-minute lessons are carried in seven periods of fifty to fifty-five minutes — so each period below runs slightly ahead of one of his.

L	Date	Wk	Content	Set for home
1	Thu 3 Sep P2 · 09:55	B	Scalars, vectors and units. Close out Energy 1 in ten minutes, then the groundwork the whole unit rests on: scalar against vector, and converting units. Put $72 \text{ km/h} \rightarrow 20 \text{ m/s}$ on the board and work it.	Conversion sheet, finish.
2	Thu 10 Sep P1 · 08:55	A	Distance and displacement, speed and velocity. Walk the route round the lab: 25 m of distance, 0 m of displacement. An object moving in a circle has constant speed and changing velocity.	Definitions and three worked examples.
3	Fri 11 Sep P1 · 08:55	A	Calculating speed. $s = vt$, rearranged both ways, with attention to layout. Then the typical everyday speeds they must know, and what changes them.	Speed calculations. Learn the typical speeds.
4	Tue 15 Sep P3 · 11:15	B	Distance–time graphs. Reading and describing motion from a graph; gradient is speed. Students act out a graph before they draw one.	AQA distance–time graphs worksheet.
5	Thu 17 Sep P2 · 09:55	B	Acceleration. $a = \Delta v/t$, units of m/s^2 , and what a negative value means. Uniform acceleration, and everyday values.	Acceleration calculations.
6	Thu 24 Sep P1 · 08:55	A	Velocity–time graphs. Gradient is acceleration; <i>area beneath the line</i> is distance travelled. The two graph types side by side — this is where marks are lost.	Mixed graph interpretation.
7	Fri 25 Sep P1 · 08:55	A	Uniform acceleration and review. $v^2 - u^2 = 2as$. Mixed motion problems, differentiated, then a review pass over the whole unit.	Revise. Test is Tuesday.

Tue 29 Sep P3 · 11:15 **Week B — UNIT TEST: Forces 1, Kinematics.** Announce it on Fri 11 Sep, a fortnight ahead; L7 is the review lesson.

Thu 1 Oct P2 · 09:55 **Week B** Return the test, then open **Forces 2** (§6.5.1, forces and their interactions).

Equations they must hold

$s = vt$	distance, speed, time
$a = \frac{\Delta v}{t}$	acceleration
$v^2 - u^2 = 2as$	uniform acceleration

Typical speeds, from memory: walking 1.5, running 3, cycling 6, car 25, train 55, aeroplane 250, sound in air $\approx 330 - 340$ — all m/s.

Held back for Year 11

The department excludes three things from this unit. They belong to *Forces 4*, the higher-content block taught after the Year 11 mocks:

- drawing a **tangent** to a curved graph
- **instantaneous** velocity or power from that tangent
- **area under a curved** graph

Introduce them if a strong group asks, but do not assess them here.

Kit, to book with Joel. L2 turntable with a blob of blu-tac (constant speed, changing velocity); L3 something worth measuring the speed of, plus tape measures; L4 and L6 PASCO MatchGraph, so they chase a graph with their own motion before interpreting one on paper. **10 P sits the same test** on its own dates — build each lesson once.